

ENVIRONMENTAL APPLICATIONS

Maximum Marks: 100

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
2. *You will **not** be allowed to write during the first 15 minutes.*
3. *This time is to be spent in reading the question paper.*
4. *The time given at the head of this Paper is the time allowed for writing the answers.*

5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
6. *The intended marks for questions or parts of questions are given in brackets[].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 11 printed pages and 1 blank page.

SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1

Choose the correct answers to the questions from the given options.

[20]

(Do not copy the questions, write the correct answers only.)

(i) Which of the following factors influence Globalisation?

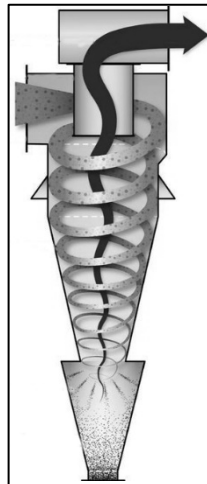
1. Debt trap
2. Freedom of trade
3. Brain drain
4. Improvements in communication

- (a) 1 and 2
- (b) 2 and 4
- (c) 3 and 4
- (d) 2 and 3

(ii) Secondary treatment of wastewater deals with:

- (a) suspended particulate matter
- (b) harmful chemicals
- (c) biological matter
- (d) solid waste

- (iii) What is the primary objective of decentralised governance?
- (a) Decisions are taken by the central government
 - (b) To eliminate the role of state government
 - (c) Bureaucratic control
 - (d) Local self-government takes all decisions
- (iv) The main reason behind generation of wasteland is:
- (a) eco-tourism
 - (b) urbanisation
 - (c) afforestation
 - (d) poor agricultural practices
- (v) The picture given below represents one of the *air control devices* used in dust collection systems in the industrial process. Identify the principle behind the device.



- (a) Gravitational force
- (b) Combustion technology
- (c) Rotation effect and gravity
- (d) Electrostatic charge

- (vi) Which of the following solutions effectively address the fuelwood crisis?
1. Using solar devices
 2. Urbanisation
 3. Afforestation
 4. Deforestation
- (a) 1 and 2
- (b) 1 and 3
- (c) 3 and 4
- (d) 1, 2 and 3
- (vii) **Assertion (A):** Waste segregation helps in reducing environmental pollution.
- Reason (R):** Segregation of waste at source allows for better recycling and disposal.
- (a) (A) is true and (R) is false.
- (b) (A) is false and (R) is true.
- (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (viii) **Assertion (A):** NGOs work for the welfare of children, women and environment.
- Reason (R):** NGOs are working for sustainable development.
- (a) (A) is true and (R) is false.
- (b) (A) is false and (R) is true.
- (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

- (ix) **Statement 1:** In Delhi, cars, buses, autos and two-wheelers are run by CNG and do not contribute to air pollution.
- Statement 2:** CNG is a non-polluting and non-toxic fuel.
- (a) Both Statement (1) and Statement (2) are true.
 - (b) Statement (1) is true and Statement (2) is false.
 - (c) Both Statement (1) and Statement (2) are false.
 - (d) Statement (1) is false and Statement (2) is true.
- (x) The main feature of Dry Compost Toilet is that it:
- (a) uses electricity
 - (b) saves water
 - (c) uses chemicals for cleaning
 - (d) increases soil pollution
- (xi) Thermal power plants use _____ to produce electricity.
- (a) coal
 - (b) uranium
 - (c) water
 - (d) the earth's internal heat
- (xii) A systematic way of analysing and formulating the rate of natural resource depletion is:
- (a) Environmental Impact Assessment
 - (b) Non-resource accounting
 - (c) Natural resource accounting
 - (d) Ecofriendly technology

- (xiii) Which of the following is a technique used by farmers to prevent soil degradation?
- (a) Mono cropping
 - (b) Overgrazing
 - (c) Mining
 - (d) Contour bunding
- (xiv) Identify the **APPROPRIATE** statements with reference to Night Soil:
- 1. Rich in nutrients
 - 2. Increases soil fertility
 - 3. Increases crop yield
 - 4. Increases pollution
- (a) 1, 2 and 4
 - (b) 2, 3 and 4
 - (c) 3, 4 and 1
 - (d) 1, 2 and 3
- (xv) What is the primary characteristic of a **micro hydel** power plant?
- (a) It uses nuclear energy.
 - (b) It requires a large dam as reservoir.
 - (c) It generates electricity up to 100 kilowatts.
 - (d) It uses fossil fuel for power generation.

- (xvi) Identify the **INCORRECT** statement with reference to biodiesel.
- (a) Gives less mileage.
 - (b) Reduces the emission of hydrocarbon.
 - (c) Not available everywhere.
 - (d) Considered to be a conventional source of energy.
- (xvii) To reduce the negative impacts on the environment, which of the following project requires EIA?
- (a) Crude oil refinery
 - (b) Paper industry
 - (c) Food industry
 - (d) Hotels and Resorts
- (xviii) Which of the following statements about community forestry is **TRUE**?
- (a) It involves JFM, Social forestry, and Agro forestry.
 - (b) It does not involve local communities.
 - (c) It discourages innovative ideas and knowledge.
 - (d) It does not affect the life of the local people.
- (xix) The agriculture sector is a valuable resource. This sector can become a Zero Percent Waste Generator by efficiently utilising its waste for:
- (a) Biogas plant, Stubble burning
 - (b) Stubble burning, Paper manufacturing
 - (c) Paper and board manufacturing, Biogas plant
 - (d) Biofuel production, Stubble burning

- (xx) Which of the following is **NOT** an indigenous method of rainwater harvesting?
- (a) Kund
 - (b) Temple pond
 - (c) Roof top rainwater harvesting
 - (d) Johad

Question 2

- (i) Give *any two* causes of soil erosion. [2]
- (ii) Mention *any two* effects of recession due to globalisation. [2]
- (iii) What is meant by watershed? Give *any two* reasons why watershed management is important. [3]
- (iv) Identify the image given below and mention *two* benefits of this structure. [3]



Question 3

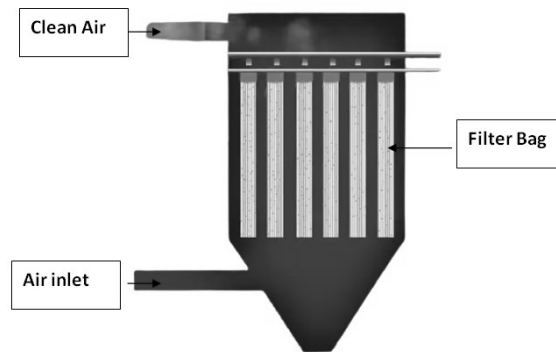
- (i) With reference to industrial waste, write *any two* differences between *effluent* and *emission*. [2]
- (ii) Give *any two* advantages of check dams. [2]
- (iii) Mention *any three* advantages of Indigenous weaving. [3]
- (iv) Discuss *any three* benefits of practising the 3Rs (Recycle, Reduce, Reuse). [3]

SECTION B (60 Marks)

(Answer *any four* questions from this *Section*.)

Question 4

- (i) Continuous unplanned deforestation is leading to many natural calamities. [5]
Explain how JFM and community forestry help to control deforestation.
- (ii) A huge number of developmental projects in a city leads to environmental [5]
degradation. It specially affects the agriculture sector because of deforestation
and soil erosion.
Describe *any five* ways how soil erosion affects food productivity.
- (iii) Identify the image given below and explain the working principle. [5]



Question 5

- (i) Curitiba Traffic Management System of Brazil is known worldwide for its [5]
innovative and sustainable traffic management system.
Explain *any five* key features of *Curitiba Traffic Management System*.
- (ii) Globalisation is imposing many positive and negative impacts on our resources. [5]
Explain *any five* impacts of globalisation on the human resource.
- (iii) TBS is an active NGO working for the benefit of people and the environment. [5]
Briefly explain *any five* key aspects of Tarun Bharat Sangh's (TBS) work.

Question 6

- (i) Explain solar energy. Give *any two* advantages and *any two* disadvantages of solar energy. [5]
- (ii) Every single step matters and can make a visible change. [5]
In reference to the above statement, each and every individual can take initiatives to restore the environment. Discuss *any five* initiatives taken by you to upgrade the environment.
- (iii) Explain *any five* reasons for recycling wastewater. [5]

Question 7

- (i) Environment Impact Assessment (EIA) is very important for the big developmental projects, to maintain balance in the ecosystem. Discuss the *two* stages and the process involved in EIA. [5]
- (ii) What are *Euro norms*? Describe *any four* limitations of implementing Euro norms in India. [5]
- (iii) Explain *any five* advantages of small dams. [5]

Question 8

- (i) Leaching of toxins from landfills is a great threat to our water bodies because toxins from the leachate also deplete our underground water. [5]
Suggest *any five* efficient methods of waste disposal which do not degrade our natural resources.
- (ii) Explain *Sustainable Development*. Describe *any three* characteristics principles of sustainability. [5]

- (iii) Promotion of small-scale industries is bringing tremendous changes in the villagers' life. [5]



With reference to the above statement, explain Dastkar.

Question 9

- (i) Economic policies play an important role in controlling air pollution. [5]
Discuss *any five* economic policies which help in controlling air pollution.
- (ii) Sapna is a social worker who works for the welfare of women. Explain *any five* [5]
ways in which Sapna can uplift the women of the community.
- (iii) Discuss *any five* modern fishing techniques which are used for commercial [5]
purposes.